Meridian Health Group — Enterprise Domain Context

About the Organization

Meridian Health Group is a regional hospital network operating **3 acute-care hospitals** and **12 outpatient clinics** across the Southeast US, serving approximately **420,000 registered patients** and processing **1.8 million appointment interactions per year**.

The **HealthConnect** digital platform is Meridian's patient- and clinician-facing technology layer. It is built as a modern cloud-native application that integrates with **Epic EHR** (version Hyperdrive 2024) as the authoritative source of record for all clinical data. HealthConnect uses **HL7 FHIR R4** as the standard integration interface to Epic.

Two product teams build on the HealthConnect platform:

- **Patient Experience Team** — patient-facing portal (web + mobile), appointments, records access, secure messaging
- **Care Coordination Team** — clinician and admin portal, referral workflows, care team management, care plans

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Enterprise User Personas

Persona	Team	Description
Patient	Patient Experience	Registered individual receiving care; self-service booking and records access
Carer / Guardian	Patient Experience	Family member managing a dependent's care; requires proxy access grant
Clinical Staff (GP / Specialist / Nurse)	Care Coordination	Manages care plans, clinical inbox, referrals, and secure messaging
Front Desk / Admin	Care Coordination	Handles appointment scheduling, patient check-in, and waitlist management
Clinical Informatics	Both	Internal analytics team; aggregate reporting; no individual-level PHI access

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Clinical Source of Record — Epic EHR

- **Epic** (Hyperdrive 2024) is the authoritative source of truth for all patient records, appointments, medications, lab results, and care plans
- **HealthConnect does not store clinical data** — it is a presentation and workflow layer that reads from and writes back to Epic via FHIR R4
- Epic FHIR resources used: `Patient`, `Appointment`, `MedicationRequest`, `DiagnosticReport`, `Observation`, `DocumentReference`, `Communication`, `CarePlan`, `Referral`
- Polling cadence: lab results every 15 minutes; radiology PDFs every 30 minutes
- Write-back: appointment cancellations, prescription requests, message threads (stored as `Communication` resources in Epic)

Company-Wide Compliance & Regulatory Framework

HIPAA (Primary)

- All **PHI (Protected Health Information)** must be encrypted at rest and in transit
- **Minimum Necessary Standard**: patients only see their own data; clinical staff see only the patients in their care
- **Audit logs required** for every PHI access event (who, what, when, source IP)
- Patient has the right to access, correct, and download their records (HIPAA Right of Access)
- **Business Associate Agreements (BAA)** required for all third-party processors (AWS, Twilio, SendGrid, Doxy.me, Stripe)
- Patient identifiers stored as MRN tokens within HealthConnect; plain MRN used only at the FHIR Adapter boundary

GDPR (EU Patients)

- EU patient data must remain in **eu-west-1** (AWS Ireland); data residency enforced at the infrastructure layer
- Right to erasure: patient may request deletion of HealthConnect account data (Epic data governed separately by clinical records laws)
- Consent management: separate opt-ins for clinical communications, research participation, and marketing

Accessibility

- WCAG 2.1 AA on all patient-facing interfaces
- Full screen reader support (JAWS and VoiceOver) tested in CI via axe-core

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Enterprise Communication Policies

- All patient-to-clinician communication is **asynchronous** (not live chat)
- Response SLA: **2 business days** for non-urgent messages; urgent escalation to on-call nurse within 15 minutes
- Messages become part of the **clinical record** (stored as FHIR `Communication` resources in Epic)
- Controlled substances cannot be requested via any digital channel — legislative requirement
- All communications respect patient preference flags (email / SMS / push); no marketing without explicit opt-in consent

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Enterprise Integration Landscape

System	Purpose	Primary Team
Epic EHR (FHIR R4)	Clinical source of record — appointments, records, medications, lab results	Both (via FHIR Adapter)
Doxy.me	Telehealth video consultation platform	Patient Experience
Twilio	SMS notifications — appointment reminders, urgent message alerts	Both
SendGrid	Transactional email — booking confirmations, lab notifications, password reset	Both

Stripe	Co-payment collection at booking (outpatient visits)	Patient Experience
Apple Health / Google Fit	Wearable health data sync (Phase 2)	Patient Experience
AWS Cognito	Patient identity and authentication	Both (shared)
AWS EventBridge	Domain event bus — appointments, messages, lab results	Both (shared)